

FAEZ BEN NAGI

B2ITE

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Questions

Answers to these questions can be found in Appendix G.

1. What is the purpose of a class definition?
2. A class has the same relation to an object that a basic data type has to a variable of that type.
3. In a class definition, data or functions designated private are accessible
 - a. to any function in the program.
 - b. only if you know the password.
 - c. to member functions of that class.
 - d. only to public members of the class.
4. Write a class definition that creates a class called leverage with one private data member, crowbar, of type int and one public function whose declaration is void pry().
5. True or false Data items in a class must be private.
6. Write a statement that defines an object called lever1 of the leverage class described in Question 4. Leverage lever1;
7. The dot operator (or class member access operator) connects the following two entities (reading from left to right):
 - a. A class member and a class object
 - b. A class object and a class
 - c. A class and a member of that class
 - d. A class object and a member of that class
8. Write a statement that executes the pry() function in the lever1 object, as described in Questions 4 and 6. Lever1.pry();
9. Member functions defined inside a class definition are inline by default. also private
10. Write a member function called getcrow() for the leverage class described in Question 4. This function should return the value of the crowbar data. Assume the function is lw

Leverage getcrow(): { return(crowbar); }
11. A constructor is executed automatically when an object is created disposed defined within the class definition.
12. A constructor's name is the same as this class of which it is a member.
13. Write a constructor that initializes to 0 the crowbar data, a member of the leverage class described in Question 4. Assume that the constructor is defined within the class definition.

Leverage() { crowbar = 0; }
14. True or false: In a class you can have more than one constructor with the same name.
15. A member function can always access the data
 - a. in the object of which it is a member.
 - b. in the class of which it is a member.
 - c. in any object of the class of which it is a member.
 - d. in the public part of its class.

class Leverage

{ private;

int crowbar;

public;

void pry();

};



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- ① What is the purpose of a class definition?
- ② A _____ has the same relation to an _____ that a basic data type has to a variable of that type.
- ③ In a class definition, data or functions designated private are accessible
 - a. to any function in the program.
 - b. only if you know the password.
 - ③ c. to member functions of that class.
 - d. only to public members of the class.
- ④ Write a class definition that creates a class called leverage with one private data member, crowbar, of type int and one public function whose declaration is void pry().
- ⑤ True or false: Data items in a class must be private.
- ⑥ Write a statement that defines an object called lever1 of the leverage class described in

Question 4.

- ⑦ The dot operator (or class member access operator) connects the following two entities (reading from left to right):
 - a. A class member and a class object
 - b. A class object and a class
 - c. A class and a member of that class
 - ⑦ d. A class object and a member of that class
- ⑧ Write a statement that executes the pry() function in the lever1 object, as described in

Questions 4 and 6.

- ⑨ Member functions defined inside a class definition are _____ by default.
- ⑩ Write a member function called getcrow() for the leverage class described in Question 4. This function should return the value of the crowbar data. Assume the function is defined within the class definition.
- ⑪ A constructor is executed automatically when an object is _____.
- ⑫ A constructor's name is the same as _____.
- ⑬ Write a constructor that initializes to 0 the crowbar data, a member of the leverage class described in Question 4. Assume that the constructor is defined within the class definition.
- ⑭ True or false: In a class you can have more than one constructor with the same name.
- ⑮ A member function can always access the data
 - a. in the object of which it is a member.
 - b. in the class of which it is a member.
 - c. in any object of the class of which it is a member.
 - d. in the public part of its class.

✓16. Assume that the member function getcrow() described in Question 10 is defined outside the class definition. Write the declaration that goes inside the class definition.

✓17. Write a revised version of the getcrow() member function from Question 10 that is defined (outside the class definition)

✓18. The only technical difference between structures and classes in C++ is that _____.

✓19. If three objects of a class are defined, how many copies of that class's data items are three. How many copies of its member functions? one

✓20. Sending a message to an object is the same as calling one of its member functions

✓21. Classes are useful because they

- a. are removed from memory when not in use.
- b. permit data to be hidden from other classes.
- c. bring together all aspects of an entity in one place.
- d. can closely model objects in the real world.

✓22. True or false: There is a simple but precise methodology for dividing a real-world programming problem into classes. false trial and error may be necessary

✓23. For the object for which it was called, a const member function,

- a. can modify both const and non-const member data.
- b. can modify only const member data.
- c. can modify only non-const member data.
- d. can modify neither const nor non-const member data.

✓24. True or false: If you declare a const object, it can only be used with const member functions.

✓25. Write a declaration (not a definition) for a const void function called aFunc() that takes

one const argument called jerry of type float void aFunc(const float jerry)
const;

* Answers to Questions

✓1. A class declaration describes how objects of a class will look when they are created.

2. class, object

3. c

4.

class leverage

{

private:

int crowbar;

public:

void pry();

};

5. false; both data and functions can be private or public

6. leverage lever1;

7. d

8. lever1.pry();
9. inline (also privat).
- 10.int getcrow()
{ return crowbar; }
11. created (defined)
12. the class of which it is a member
13.
leverage()
{ crowbar = 0; }
14. true
15. a
16. int getcrow();
17.
int leverage::getcrow()
{ return crowbar; }
18. member functions and data are, by default, public in structures but private in classes
19. three, one
20. calling one of its member functions
21. b, c, d
22. false; trial and error may be necessary
23. d
24. true
25. void aFunc(const float jerry) const;

Questions

Answers to these questions can be found in Appendix G.

1. Inheritance is a way to
 - a. make general classes into more specific classes.
 - b. pass arguments to objects of classes.
 - c. add features to existing classes without rewriting them.
 - d. improve data hiding and encapsulation. = OOP
2. A "child" class is said to be derived from a base class.
3. Advantages of inheritance include
 - a. providing class growth through natural selection.
 - b. facilitating class libraries. إنشاء المكتبة العامة inheritance
 - c. avoiding the rewriting of code. إعادة كتابة
 - d. providing a useful conceptual framework. →
4. Write the first line of the specifier for a class Bosworth that is publicly derived from a class Alphonso.
class Bosworth: public Alphonso
5. True or false? Adding a derived class to a base class requires fundamental changes to the base class.
6. To be accessed from a member function of the derived class, data or functions in the base class must be public or protected. protected
7. If a base class contains a member function basefunc(), and a derived class does not contain a function with this name, can an object of the derived class access basefunc()? Yes. ⇒ overriding ⇒ base class overriding
8. Assume that the classes mentioned in Question 4 and the class Alphonso contain a member function called alfunc(). Write a statement that allows object BosworthObj of class Bosworth to access alfunc().
BosworthObj.alfunc() constructor without arguments
9. True or false: If no constructors are specified for a derived class, objects of the derived class will use the constructors in the base class.
10. If a base class and a derived class each include a member function with the same name, ⇒ overriding مفوض which member function will be called by an object of the derived class, assuming the scope-resolution operator is not used? ⇒ one in the derived class. الاولوية الى في الاصناف
11. Write a declarator for a no-argument constructor of the derived class Bosworth of Question 4 that calls a no-argument constructor in the base class Alphonso.
Bosworth(): Alphonso() connect constructor with constructor
12. The scope-resolution operator usually
 - a. limits the visibility of variables to a certain function.
 - b. tells what base class a class is derived from. من الوراثة
 - c. specifies a particular class. توضيح كلاس جزئية
 - d. resolves ambiguities. نتيجة للغموض Errors
13. True or false: It is sometimes useful to specify a class from which no objects will ever be created.

هل يجب ان يكون دائماً انه
 مفوض كلاس =

created.

14. Assume that there is a class Derv that is derived from a base class Base. Write the declarator for a derived-class constructor that takes one argument and passes this argument Base (arg) along to the constructor in the base class.

15. Assume a class Derv that is privately derived from class Base. An object of class Derv

located in main() can access

- a. public members of Derv.
- b. protected members of Derv. x
- c. private members of Derv. x
- d. public members of Base.
- e. protected members of Base.
- f. private members of Base.

16. True or false: A class D can be derived from a class C, which is derived from a class B, which is derived from a class A. "levels"

17. A class hierarchy

- a. shows the same relationships as an organization chart.
- b. describes "has a" relationships.
- c. describes "is a kind of" relationships.
- d. shows the same relationships as a family tree.

18. Write the first line of a specifier for a class Tire that is derived from class Wheel and from class Rubber. Multi-class Tire: public Wheel, public Rubber

19. Assume a class Derv derived from a base class Base. Both classes contain a member function func() that takes no arguments. Write a statement to go in a member function

of Derv that calls func() in the base class. Base::func();

20. True or false: It is illegal to make objects of one class members of another class.

21. In the UML, inheritance is called generalization

22. Aggregation is

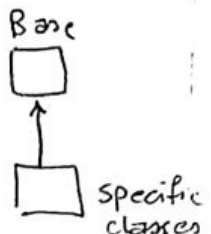
- a. a stronger form of instantiation.
- b. a stronger form of generalization.
- c. a stronger form of composition.
- d. a "has a" relationship.

23. True or false: the arrow representing generalization points to the more specific class.

24. Composition is a stronger form of aggregation

Answers to Questions

- 1. a, c
- 2. derived
- 3. b, c, d
- 4. class Bosworth : public Alphonso
- 5. false
- 6. protected
- 7. yes (assuming basefunc is not private)



الإجابات

8. `BosworthObj.alfunc();`
9. true
10. the one in the derived class
11. `Bosworth() : Alphonso() { }`
12. c, d
13. true
14. `Derv(int arg) : Base(arg)`
15. a
16. true
17. c
18. `class Tire : public Wheel, public Rubber`
19. `Base::func();`
20. false
21. generalization
22. d
23. false
24. stronger, aggregation.